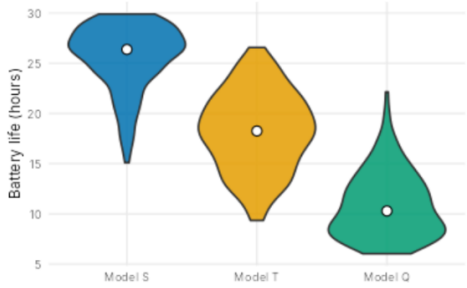
DSA06 - Data Handling and Visualization | Activity: Identify Skewness using Violin Plots | Slot D

R.No: 18 Reg. No: 192325036 Name: KUMMARI HEMA SAI SIDDARTHA

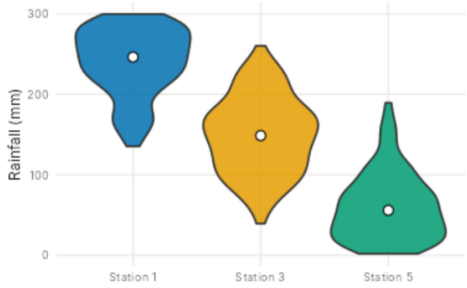
Part A (18 marks). Each chart is a violin plot (white dot = median) made in R with ggplot2. For every violin circle the skew (1 mark) and the relation of the mean to the median (0.5 mark).

Chart 1: Battery life (hours)



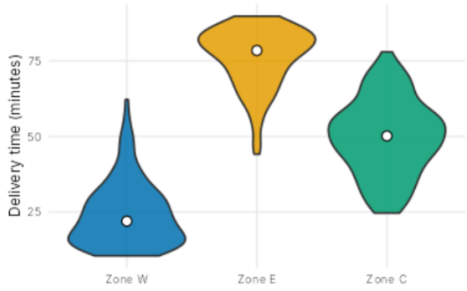
Violin	Skew (circle one)	Mean vs Median (circle)
Model S	Right / Left / Symm.	Mean > / < / ~ Median
Model T	Right / Left / Symm.	Mean > / < / ~ Median
Model Q	Right / Left / Symm.	Mean > / < / ~ Median

Chart 2: Rainfall (mm)



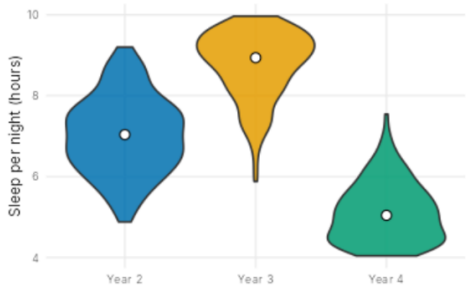
Violin	Skew (circle one)	Mean vs Median (circle)
Station 1	Right / Left / Symm.	Mean > / < / ~ Median
Station 3	Right / Left / Symm.	Mean > / < / ~ Median
Station 5	Right / Left / Symm.	Mean > / < / ~ Median

Chart 3: Delivery time (minutes)



Violin	Skew (circle one)	Mean vs Median (circle)
Zone W	Right / Left / Symm.	Mean > / < / ~ Median
Zone E	Right / Left / Symm.	Mean > / < / ~ Median
Zone C	Right / Left / Symm.	Mean > / < / ~ Median

Chart 4: Sleep per night (hours)



Violin	Skew (circle one)	Mean vs Median (circle)
Year 2	Right / Left / Symm.	Mean > / < / ~ Median
Year 3	Right / Left / Symm.	Mean > / < / ~ Median
Year 4	Right / Left / Symm.	Mean > / < / ~ Median

Part B (2 marks)

B1. In Chart 4, violin "Year 3" is not symmetric. Towards which side is its tail, and which is the better measure of a typical sleep per night here: mean or median? Give one reason. [1]

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1]

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DSA06 – Data Handling and Visualization

Part A: Identify Skewness Using Violin Plots

In a violin plot, the white circle represents the median. The shape of the violin indicates the distribution and its tail.

- Right-skewed: The tail extends towards the higher values; generally, mean > median.
- Left-skewed: The tail extends towards the lower values; generally, mean < median.
- Symmetric: Both sides are approximately balanced; generally, mean ≈ median.

Chart 1: Battery Life (hours)

Violin	Skew	Mean vs Median
Model S	Left-skewed	Mean < Median
Model T	Approximately symmetric	Mean ≈ Median
Model Q	Right-skewed	Mean > Median

Chart 2: Rainfall (mm)

Violin	Skew	Mean vs Median
Station 1	Left-skewed	Mean < Median
Station 3	Approximately symmetric	Mean ≈ Median
Station 5	Right-skewed	Mean > Median

Chart 3: Delivery Time (minutes)

Violin	Skew	Mean vs Median
Zone W	Right-skewed	Mean > Median
Zone E	Left-skewed	Mean < Median
Zone C	Approximately symmetric	Mean ≈ Median

Chart 4: Sleep per Night (hours)

Violin	Skew	Mean vs Median
Year 2	Approximately symmetric	Mean ≈ Median
Year 3	Left-skewed	Mean < Median
Year 4	Right-skewed	Mean > Median

Note: These classifications are based on the visible violin shapes. Some distributions may be only approximately symmetric.

Part B

B1. Chart 4 – Year 3

Answer: The tail extends towards the left (lower sleep hours), so it is left-skewed. The median is the better measure of typical sleep because it is less affected by extreme values.

B2. R code using ggplot2

```
ggplot(df, aes(x = group, y = value)) +
```

```
  geom_violin() +
  stat_summary(fun = median, geom = "point",
               color = "red", size = 3)
```

This code draws a violin plot for each group and displays the median as a red dot.